

Hands-On or Not? Comparing Mid-Air and Passive Haptics in Button-Based VR Training Tasks

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ABSTRACT

This pilot study investigates the impact of passive haptics in VR training with simple control elements. 26 participants completed tasks using either mid-air interaction or a physical mock-up providing passive haptic feedback. Learning outcomes were assessed via practical transfer and knowledge tests, and user experience was measured through presence, workload, and usability ratings. No significant differences were found between interaction conditions across any measure. These findings provide initial evidence that, for brief and discrete control-element interactions, passive haptics do not substantially affect performance or experience, guiding future design decisions and studies involving more complex VR training tasks.

Index Terms: Virtual reality, passive haptic, training.

1 INTRODUCTION

In various contexts such as industrial, medical, and military training, personnel must learn to operate systems that rely on the correct use of specific tools, machines, and control interfaces. Virtual reality (VR) has been widely explored as an alternative for this type of training, as it allows realistic scenarios to be simulated within controlled and repeatable environments [4]. However, prior work has shown that VR-based training does not always lead to learning outcomes comparable to those achieved through training with real physical systems, particularly when interaction with virtual elements differs from real-world operation [5].

According to Hwang et al [5], one factor that may contribute to these limitations is the reduced availability of sensory cues during interaction. VR interaction often relies predominantly on visual feedback, while other modalities that are relevant for skill acquisition, such as haptic feedback, are either simplified or absent. Haptic information has been shown to play an important role in the learning and transfer of manual skills, particularly in tasks involving physical manipulation or tool use [5]. Consequently, haptic feedback has been widely studied in VR training scenarios such as laboratory procedures [3], first aid training [1], and industrial or military applications [6, 9].

Despite this growing body of work, most existing studies focus on interactions that involve grasping, manipulating, or handling physical objects [1, 3, 6]. In contrast, many real-world training scenarios rely on simpler interaction primitives, such as pressing buttons or interacting with control panels, where the physical action is brief and constrained, but correctness is critical. While some prior work has explored button-based interaction using pseudo-haptic techniques [8, 2], there is still limited empirical evidence clarifying whether the integration of passive haptics through physical mock-ups leads to measurable differences in learning outcomes or user



Figure 1: Participant interacting with the physical control panel during the passive haptics condition (B) in the training task, using a Meta Quest 3 and a Unity-based VR application (OculusXR Plugin 1.9.1).

experience compared to mid-air interaction.

To address this gap, we present a pilot study that investigates the impact of passive haptics in simple button-based VR training interactions. We compare mid-air interaction with interaction supported by a calibrated physical mock-up of a control panel (Figure 1), examining potential differences in learning transfer, functional knowledge acquisition, and subjective experience. As an exploratory evaluation, this study considers both objective performance in a practical transfer task and subjective measures of presence, perceived workload, and usability focused on button interaction. The results of this pilot study aim to clarify whether integrating passive haptics into simple button-based VR training interactions leads to observable differences in learning outcomes and user experience, providing initial evidence to support design decisions in interface-based training scenarios.

2 METHOD

We conducted a mixed experimental study comparing two VR training conditions: mid-air interaction (Condition A) and VR calibrated with a physical mock-up providing passive haptic feedback (Condition B). Interaction condition was treated as a within-subject factor, as all participants experienced both conditions. To control for potential sequence and learning effects, condition order (A→B vs. B→A) was counterbalanced across participants and modeled as a between-subject factor.

2.1 Apparatus and Materials

The VR application simulated the engineering control room of a ship, visually presenting the full environment while restricting interaction to a specific control panel section. This panel contained 55 buttons, from which 10 buttons were selected for the study.

These buttons were systematically divided into two matched sets (S1 and S2), each comprising five elements carefully chosen to ensure balance in functional difficulty and spatial location, thereby avoiding bias toward either interaction condition. To further mitigate learning effects, the interaction order within each set was fixed but counterbalanced across four predefined sequences. Participants were assigned to one of two counterbalanced groups:

- Group 1 (G1): Condition A + S1 → Condition B + S2
- Group 2 (G2): Condition B + S1 → Condition A + S2

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Table 1: Comparison of learning and subjective measures between interaction conditions.

Measure	Mid-Air (A) $\mu \pm \sigma$	Passive Haptics (B) $\mu \pm \sigma$	Δ (B-A)	Test	p	Effect size
Transfer score (0-5)	4.29 $\pm$ 0.71	4.45 $\pm$ 0.70	+0.16	t-test	0.261	0.226 (d)
Knowledge score (0-5)	4.54 $\pm$ 0.90	4.54 $\pm$ 0.65	0.00	Wilcoxon	0.89	0.027 (r)
Presence	94.04 $\pm$ 6.74	95.15 $\pm$ 7.43	+1.12	t-test	0.49	0.136 (d)
NASA-TLX	15.12 $\pm$ 7.00	15.00 $\pm$ 6.34	-0.12	t-test	0.89	-0.028 (d)
Button usability (1-7)	6.23 $\pm$ 0.82	5.58 $\pm$ 1.68	-0.65	Wilcoxon	0.13	0.293 (r)

2.2 Measures and Tasks

Each participant completed a pre-test collecting demographic information and prior experience, followed by two VR training sessions, one per interaction condition. After each session, participants completed a practical transfer task, a knowledge assessment, and subjective evaluations including presence (Witmer & Singer), workload (NASA-TLX), and, for Condition B only, selected HXI [7] items assessing realism (RL), harmony (HR), involvement (IN), and autotelic (AT) experience. The HXI was used informatively to characterize the experience with passive haptics mock-up rather than as a comparative metric. Given the simplicity of the task, the passive and physically aligned haptic feedback, and the low likelihood of sensory mismatches, the Discord dimension and overall HXI score were omitted to reduce participant burden in the within-subjects protocol.

2.2.1 VR Training Task (VTT)

During each condition, participants interacted with five control elements from the corresponding set (S1 or S2). For each element, participants were instructed to locate the control, activate it, and read aloud the system feedback displayed in VR, ensuring engagement with the functional meaning of each control and supporting the training objective.

2.2.2 Practical Transfer Task

After each VTT, participants completed a practical transfer task using the physical control panel mock-up. Participants were presented with operational questions and asked to identify which control they would activate. Functional learning transfer was scored based on the number of attempts required to select the correct control (1st attempt=1.0, 2nd=0.5, 3rd=0.25, ≥ 4 attempts/no response=0).

2.2.3 Learning Assessment

Learning was assessed using domain-specific knowledge questions administered after each interaction condition, immediately following the practical transfer task (post-tests). The questions targeted the functional meaning and operational use of the trained controls and included an "I do not know" option to discourage guessing.

3 RESULTS AND DISCUSSION

A total of 26 participants ($n=26$; 19 male, 7 female) were included in this study, equally distributed into two groups ($G1=13$, $G2=13$). Ages ranged from 19-45 years, with the majority (80.7%) between 19-25. All participants reported prior experience with virtual reality, and most (76.9%) were familiar with driving a car, which is relevant given that the VR control panel involved concepts analogous to vehicle operation (e.g., clutch and engine control). Although none of the participants had prior experience in a naval context, they achieved high performance overall, with a mean transfer score of 4.38 out of 5 and a mean knowledge score of 4.54 out of 5, indicating effective learning through the VR training application. In Condition B, participants reported high haptic experience ($HR=6.04$, $RL=5.90$, $IN=5.65$, $AT=6.50$; 7-point scale), indicating tactile feedback was perceived as realistic, engaging, and enjoyable.

As summarized in Table 1, no statistically significant differences were observed between Condition A and Condition B across learning, presence, workload, or usability measures. Learning transfer and knowledge acquisition were comparable between conditions, with only small, non-significant differences in mean scores and consistent small effect sizes. Subjective measures followed a similar pattern: presence and perceived workload remained stable across interaction modalities, suggesting that the availability

of passive haptic cues did not substantially alter the overall training experience for this task. While usability ratings showed a non-significant trend favoring mid-air interaction, the results collectively suggest that, for simple and discrete button-based interactions, integrating passive haptics does not provide clear advantages over mid-air interaction. These findings support the interpretation that the benefits of passive haptics in VR training may be task-dependent and less pronounced when interaction is limited to brief, low-complexity actions such as button pressing. We acknowledge several limitations of this pilot study, including the small sample size, the use of non-expert participants, and the simplicity of the task. Future work will address these limitations by increasing sample size, including naval cadets, and evaluating more complex training exercises with the control panel, such as responding to simulated engine emergencies.

4 CONCLUSION

This pilot study indicates that, for simple button-based VR interactions, there are no significant differences in learning outcomes, presence, workload, or usability between passive haptics and mid-air interaction. Consistently small effect sizes suggest that any differences were minor in practical terms for this specific task. These findings provide initial evidence to guide design decisions in VR interface-based training. Future work will extend this evaluation to more complex tasks and expert participants to determine whether these results generalize to scenarios requiring higher cognitive and motor engagement, thereby addressing the identified limitations.

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